
Synthetic Cultural Agents 2.0: Beyond Persona-Prompting

A Position Paper on Structural Preference Estimation for Culturally Aligned Language Models

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August 2026 · *Working document — position paper & empirical results*

Status. Intermediate summary of SCA 2.0: structural conjecture, implemented DPO pipeline, Tier-1 synthetic pair recovery, and Tier-2 out-of-sample empirical evaluation on World Values Survey Wave 7 items ($N = 35$). Code & data: github.com/EconLLM-Lab/SCA2_PofW.

Abstract

Large language models are increasingly used as simulators of human judgment in economics and adjacent social sciences, yet their default alignment reflects predominantly Western, Educated, Industrialized, Rich, and Democratic (WEIRD) preference data. Earlier Synthetic Cultural Agents (SCA 1.0) addressed this gap by conditioning frontier models on ethnographic profiles at inference time. That approach is useful for piloting but is reduced-form: cultural signal lives only in the prompt, is fragile to model updates, and does not support formal identification or counterfactual analysis.

This position paper outlines SCA 2.0, a structural alternative. We treat Direct Preference Optimization (DPO) as maximum-likelihood estimation in a Bradley–Terry random-utility model, using the Global Preferences Survey (GPS) six-dimensional cultural state vector to generate culture-conditioned synthetic preference data. A student model (Llama-3.1-8B-Instruct) is fine-tuned via QLoRA with unconditioned prompts (`PROMPT_COUNTRY_CONDITIONING = False`) so that cultural dispositions are encoded into parameters rather than transient context.

We evaluate the framework across two tiers: (i) Tier-1 synthetic pair recovery ($n = 200$), where own-country adapters achieve 70.0–78.5% accuracy; and (ii) Tier-2 empirical out-of-sample evaluation on $N = 35$ unseen World Values Survey (WVS) Wave 7 items (30 mapped to the six GPS dimensions; 5 demographic items). On non-WEIRD targets (Mexico), the matched MEX adapter’s fixed response distribution lies closer to Mexico’s observed responses than the USA adapter’s on 97.1% of questions ($\Delta\text{TVD} = 0.1495$, 95% CI: $[0.0964, 0.2152]$); with unconditioned prompts this difference can only arise from the adapter weights, and it establishes distributional alignment rather than country-conditioned inference. Conversely, on US targets the USA adapter degrades fidelity to the WVS response distributions relative to the base model; the mechanism (e.g., a pretraining-prior interaction versus overfitting to the synthetic scenario format) is not identified by this design. Option log-likelihood scoring exhibits systematic under-dispersion relative to human populations. We formalize the measurement argument through a Cronbach–Meehl nomological network linking the GPS source instrument, the DPO training domain, and the WVS criterion domain, and we delineate reliable applications (relative comparative statics, pre-field instrument stress-testing) from unreliable ones (absolute headcount polling, fine-tuning over already-dominant priors).

Keywords: cultural preferences; Direct Preference Optimization; Bradley–Terry; Global Preferences Survey; structural estimation; construct validity; World Values Survey

1. Introduction

Social scientists increasingly use large language models (LLMs) as low-cost simulators of survey responses, experimental subjects, and policy interlocutors. The practical appeal is clear: a model can generate thousands of responses in minutes, enabling exploratory design work before expensive fieldwork. The scientific risk is equally clear. Preference fine-tuning and human feedback data for frontier models are concentrated in WEIRD populations [?, ?]. When a model is asked to “answer as someone in Lagos, Buenos Aires, or Mexico City,” its behavior still largely reflects that default alignment—unless culture is introduced in a disciplined way.

The first generation of Synthetic Cultural Agents [?] showed that persona prompting—conditioning a frontier model on demographic and ethnographic profiles—can reproduce non-WEIRD patterns in economic experiments to a surprising degree. For piloting and hypothesis generation, that result is valuable. Methodologically, however, persona prompting is a reduced-form intervention. Cultural information lives in the context window; it vanishes when the session ends; and it provides no structural object (preferences, technology, information) that can be identified, tested, or counterfactually varied.

SCA 2.0 reframes the problem as structural preference estimation. The core conjecture is:

Cultural preferences measured by the Global Preferences Survey can be encoded into language-model policies via preference fine-tuning, so that implied utilities recover the direction of culture-conditioned choices and support formal evaluation against independent survey moments.

Three features distinguish the program from prompt engineering. First, culture is parameterized by an externally measured vector—the six GPS dimensions of Falk et al. [?]¹—rather than by free-form narrative alone. Second, training uses Direct Preference Optimization [?], which is mathematically a reparameterization of Bradley–Terry maximum likelihood [?, ?]; this licenses econometric language (utility, identification, overidentification) with minimal translation cost. Third, validation is hierarchical: GPS-targeted synthetic pairs identify the cultural state; World Values Survey (WVS) items are reserved for out-of-sample evaluation; held-out behavioral tasks assess generalization.

This document serves as both a position paper and a report on completed out-of-sample empirical evaluations on WVS Wave 7 data. Its goals are to (i) state the conjecture and design cleanly, (ii) present an honest reading of Tier-1 and Tier-2 empirical evaluations, (iii) frame construct validity for latent cultural variables in LLMs, and (iv) demarcate reliable applications from invalid downstream uses.

2. Related Work

Three short literatures motivate the design.

The GPS [?] supplies incentivized measures of patience, risk taking, positive and negative reciprocity, altruism, and trust across 76 countries—a low-dimensional, cross-culturally comparable parameter vector. Country means are not individuals (within-country variance is often large), but they are a defensible starting point for population-level simulation. A growing experimental literature uses LLMs as simulated respondents [?, ?]; much of it is reduced-form role prompting, the family to which SCA 1.0 belongs. Complementary work documents cultural gaps in model outputs [?, ?] but typically stops at evaluation rather than estimation. DPO [?] optimizes a policy so preferred responses receive higher probability than dispreferred ones relative to a reference model; under standard assumptions this coincides with Bradley–Terry likelihood for a scaled log-probability score—the bridge to structural econometrics.

What is missing—and what SCA 2.0 targets—is a pipeline that grounds preference data in an external cultural instrument, fine-tunes open-weight models rather than relying on prompts alone, and subjects the resulting agents to rigorous out-of-sample evaluation.

3. Methodology

3.1 Primitives

Let x denote a decision scenario (prompt) and y a free-text response. A cultural state is a vector

$$z = (\delta, \gamma, \xi^+, \xi^-, \alpha, \tau)^\top \in \mathbb{R}^6, \quad (1)$$

corresponding to time preference (patience), risk taking, positive reciprocity, negative reciprocity, altruism, and trust from the GPS. An SCA is a policy $\pi(\cdot | x; z)$ whose behavior is governed by a separable reward

$$R(x, y; z) = r_{\text{task}}(x, y) + \phi(z)^\top m(x, y), \quad (2)$$

where r_{task} captures generic competence (coherence, relevance), $m(x, y) \in \mathbb{R}^6$ is a vector of measurable behavioral features of the response, and $\phi : \mathbb{R}^6 \rightarrow \mathbb{R}^6$ maps GPS scores into reward weights.

The structural assumption. Equation (??) is a *structural* (not descriptive) assumption: it claims that the country-specific component of an agent’s behavior is a linear function of six externally measured preference traits, mediated by observable features of the response. Two testable implications follow. First, *separability*: the task term r_{task} is country-invariant, so any cross-country behavioral difference must load on $\phi(z)^\top m(x, y)$. Second, *dimensional structure*: only six dimensions of cultural variation are permitted to matter. Both implications are falsifiable—the first by the matched-versus-cross specificity design (Section 4), the second by out-of-sample items that “should not” be explained by any dimension and by items whose loading is measured rather than imposed.

3.2 Choice model

We treat the agent’s response as a discrete choice over a finite option set $\mathcal{Y} = \{y_1, \dots, y_K\}$ (the response codes of a survey item or the two completions of a preference pair). A Luce–Bradley–Terry–McFadden random-utility model sets

$$\Pr(y_k | x; z) = \frac{\exp(R(x, y_k; z))}{\sum_{j=1}^K \exp(R(x, y_j; z))}, \quad (3)$$

which is the softmax over option rewards. For a pair $\{y^w, y^\ell\}$, equation (??) collapses to the familiar logistic paired-comparison probability

$$\Pr(y^w \succ y^\ell | x) = \sigma(R(x, y^w; z) - R(x, y^\ell; z)), \quad (4)$$

with $\sigma(u) = (1 + e^{-u})^{-1}$.

Equation (??) is the exact object the evaluation pipeline estimates: the option-likelihood protocol (Section 4) computes $\hat{p}_k \propto \exp(\log\text{prob}(y_k | x))$ over all valid response codes and compares the resulting distribution to the empirical population distribution. The LLM is therefore used as a *choice-probability simulator*, not a text generator—a deliberate design choice that makes the model’s output directly comparable to the random-utility object in equation (??).

3.3 DPO as structural estimation

Given preference data $\{(x_i, y_i^w, y_i^\ell)\}_{i=1}^n$ in which y^w is preferred to y^ℓ under cultural state z , the Bradley–Terry likelihood is

$$\mathcal{L}(R) = \prod_{i=1}^n \sigma(R(x_i, y_i^w; z) - R(x_i, y_i^\ell; z)). \quad (5)$$

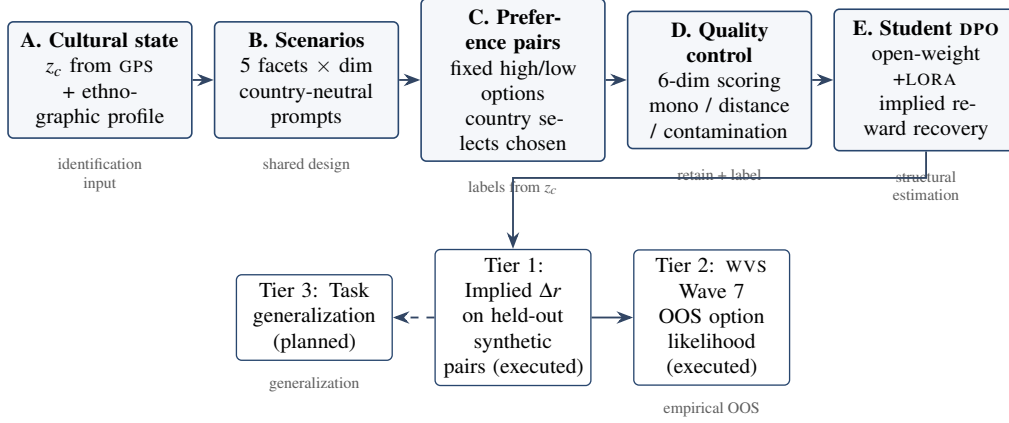


Figure 1: SCA 2.0 pipeline architecture. Solid lines mark executed pipeline stages. Culture enters as an external GPS vector z_c ; synthetic pairs are generated, quality-controlled, and fine-tuned via DPO; model validation proceeds through Tier-1 synthetic pair recovery and Tier-2 empirical WVS Wave 7 option-likelihood evaluation.

Direct Preference Optimization [?] maximizes this likelihood while reparameterizing the reward difference through the log-ratio of the trained policy to a frozen reference policy:

$$R(x, y_1; z) - R(x, y_2; z) = \beta \left[\log \frac{\pi_\theta(y_1 | x)}{\pi_{\text{ref}}(y_1 | x)} - \log \frac{\pi_\theta(y_2 | x)}{\pi_{\text{ref}}(y_2 | x)} \right], \quad (6)$$

where π_{ref} is a frozen reference model and $\beta > 0$ is a scale temperature. Equation (??) is the empirical analogue of the structural reward in equation (??): after training, the analyst can compute, for any held-out pair, the *implied reward margin* $\widehat{\Delta r} = \beta [\log\text{-ratio of chosen} - \log\text{-ratio of rejected}]$, which is the estimated utility difference under the country-specific z .

Identification and its limits. Equation (??) identifies reward *differences* up to scale (β) and location (any additive constant cancels in paired comparisons). The two structural features we *can* recover are (i) the sign of Δr —which option a country’s policy ranks higher—and (ii) the own-versus-other contrast, i.e., whether an adapter trained on country c ’s labels assigns systematically larger margins to c ’s chosen options than an adapter trained on country c' . We cannot recover the absolute level of R nor, in general, the individual coordinates of $\phi(z)$ without additional moment conditions—a limitation we make explicit rather than obscure.

3.4 Pipeline Architecture

Figure ?? summarizes the pipeline architecture across data generation, fine-tuning, and out-of-sample validation.

3.5 Construct validity and the nomological network

Because z is a *latent* construct—nobody observes “patience in Mexico” directly—the scientific status of the pipeline hinges on construct validity. We adopt the classical formulation of Cronbach and Meehl [?]: a construct is validated by embedding it in a *nomological network*, a set of theoretical relations linking the construct to (a) other constructs and (b) observable measures. Validation is then a matter of testing whether the observed correlations behave as the network predicts.

The network. For the latent construct “country-level economic preference disposition,” the network has three nodes—three families of observables that the construct must systematically relate to:

1. **Source domain (identification):** the GPS instrument z_c (Falk et al., 2018), an incentivized, cross-nationally comparable measure of the six preference dimensions. This is the *defining* measure of the construct: z_c is what we mean by “Mexico’s preference profile” in this paper.
2. **Training domain (internal manifestation):** the synthetic preference pairs $\{(x_i, y_i^w, y_i^\ell)\}$ whose labels are generated by z_c , and the implied reward margins $\widehat{\Delta r}$ recovered from the trained policy.
3. **Criterion domain (external manifestation):** the World Values Survey (WVS) item-response distributions P_{WVS} , an independent, human-administered survey that was *not* used in training and whose items are mapped to the six dimensions via a curated schema (`question_map_wvs_edited.csv`).

The network postulates three links, each with a distinct falsifiable prediction:

1. **Link 1 (source $\rightarrow$ training):** preferences implied by z_c must rank y^w above y^ℓ on held-out pairs. Falsified if the trained policy fails to recover its own labels (Tier-1 preference accuracy ≤ 0.5).
2. **Link 2 (training $\rightarrow$ criterion, convergent validity):** an adapter trained on z_c must reduce distributional error against P_{WVS} for country c on unseen items, relative to the base model—because both the synthetic labels and the WVS responses are driven by the same underlying z_c . Falsified if fine-tuning does not improve (or degrades) empirical fit.
3. **Link 3 (criterion-domain discrimination, discriminant validity):** an adapter trained on z_c must fit c ’s WVS distributions better than an adapter trained on $z_{c'}$, on the *same* items, with country names withheld from the prompt. Falsified if the matched adapter does not dominate the cross adapter (matched-vs-cross specificity ≤ 0.5).

How the network handles negative evidence. The network is a *joint* claim, and its links can fail independently—which is precisely what the data show. The USA adapter *passes* Link 1 (Tier-1 accuracy 0.70) but *fails* Links 2–3 on US WVS items: the base model already encodes a strong US pretraining prior, so the adapter’s additional synthetic signal over-corrects rather than refines. This is not evidence against the construct; it is evidence that the *source domain* is redundant with the model’s pretraining for WEIRD populations. The same network then correctly predicts that the MEX adapter—trained on a population outside the pretraining prior—should pass all three links, which it does on the aggregate Tier-2 metrics (Tier-1 accuracy 0.785; matched-better fraction 97.1% on MEX items; dimension-level convergence such as Altruism TVD falling from 0.450 to 0.152). Two caveats are warranted. First, matched-vs-cross compares each adapter’s *fixed* output distribution against each country’s observed responses—the underlying option probabilities are country-invariant by design, so the claim is about distributional alignment, not inference-time conditioning. Second, on Mexico items the improvement relative to the *base* model is concentrated in altruism, patience, and positive reciprocity; trust and risk-taking items improve relative to the cross adapter but not relative to the base model.

Reliability. Reliability—consistency of measurement—is assessed within the same framework: bootstrap confidence intervals over questions (2,000 replications) for every reported metric; calibration tables (predicted versus empirical option frequencies, ECE); and signed moment diagnostics (mean and dispersion bias) that reveal the structural under-dispersion of softmax option scoring. These diagnostics do not merely qualify the results; they are part of the nomological network, because they bound how much of the observed fit is construct signal versus scoring artifact.

4. Empirical Results

4.1 Tier-1: Synthetic Preference Recovery

Table ?? reports Tier-1 results on held-out synthetic preference pairs ($n = 200$ per adapter–country cell). Both the Mexico and US adapters recover their own culture-conditioned synthetic labels well above chance (78.5% and 70.0% preference accuracy, respectively), while cross-country accuracy remains near chance (52.0% and 45.5%).

Table 1: Tier-1 Cross-evaluation of country adapters on held-out synthetic preference pairs ($n = 200$ per cell). Preference accuracy is the share of pairs with positive implied reward margin ($\Delta r > 0$).

Adapter	Eval Data	n	Mean Δr	Pref. Accuracy	Mean $P(\text{chosen})$
Mexico Adapter	Mexico	200	+1.70	0.785	0.714
Mexico Adapter	United States	200	−0.22	0.520	0.501
USA Adapter	Mexico	200	−0.20	0.455	0.464
USA Adapter	United States	200	+0.78	0.700	0.632

4.2 Tier-2: World Values Survey Out-of-Sample Evaluation

Tier-2 tests frozen adapters on $N = 35$ unseen Wave 7 WVS survey questions under unconditioned prompts (PROMPT_COUNTRY_CONDITIONING = False). Generated option distributions $\hat{P}$ are evaluated against survey-weighted human response distributions P_{WVS} . Table ?? summarizes the primary evaluation metrics.

Table 2: Tier-2 Out-of-Sample Evaluation on $N = 35$ unseen WVS Wave 7 survey questions under unconditioned prompts. Matched vs Cross (% Matched Better) reports the fraction of questions where the matched adapter achieves lower error than the cross adapter.

Target Population	Model / Adapter	TVD ↓	JSD ↓	ECE ↓	Matched vs Cross (% Matched Better)
Mexico Data (MEX WVS)	Base Model (Llama-3.1-8B)	0.5211	0.2165	0.0919	—
Mexico Data (MEX WVS)	MEX Adapter (Matched)	0.4882	0.2010	0.0967	97.1% (34/35)
Mexico Data (MEX WVS)	USA Adapter (Cross)	0.6377	0.2965	0.1220	—
USA Data (USA WVS)	Base Model (Llama-3.1-8B)	0.3958	0.1439	0.0654	—
USA Data (USA WVS)	MEX Adapter (Cross)	0.3888	0.1405	0.0719	—
USA Data (USA WVS)	USA Adapter (Matched)	0.5258	0.2246	0.0958	11.4% (4/35)

Key Finding 1: Distributional Alignment on Non-WEIRD Targets. On Mexico WVS data, the matched MEX adapter achieves strong distributional alignment, outperforming the cross USA adapter on 97.1% of questions (34/35 items). The mean difference in Total Variation Distance ($\text{TVD}_{\text{cross}} - \text{TVD}_{\text{matched}}$) is +0.1495 (95% CI: [0.0964, 0.2152]). Because prompts omit country names, this difference can only arise from the adapter weights: each adapter emits a single fixed response distribution, and the MEX adapter’s fixed distribution sits closer to Mexico’s observed WVS responses than the USA adapter’s. This is evidence that country-specific dispositional signal was internalized into the weights during fine-tuning; it does not demonstrate that an adapter conditions on or “knows” its target country at inference time.

Plain-language reading. If the model were answering the WVS in the same way regardless of which adapter was loaded—i.e., if the adapter weights were doing nothing—we would expect the matched and cross adapters to perform equally well on any given question, and the split between them would be roughly 50/50. A win rate of 34 out of 35 (97.1%) is far from that benchmark, with a confidence interval that does not cross zero. The economic content of this finding is that fine-tuning has shifted the model’s implicit “default person answering the survey” toward one whose disposition profile matches Mexico’s, and this shift is detectable in the marginal choice probabilities of structured survey items that were *never seen during training*. The result is directional, not calibrated: we are not claiming that the adapter predicts the percentage of Mexicans who

trust their family, only that it ranks trust patterns in the direction the empirical distribution supports.

Key Finding 2: USA Adapter Degradation (mechanism unresolved). On USA WVS data, the matched USA adapter performs *worse* than both the base model (TVD = 0.5258 vs. 0.3958) and the cross MEX adapter (TVD = 0.3888). Only 11.4% of questions favored the matched USA adapter. The degradation is not confined to US data: the USA adapter also worsens fit on Mexico’s items (TVD = 0.6377 vs. base 0.5211). The mechanism—for example, a collision with the base model’s strong American pretraining prior, versus overfitting to the synthetic scenario format—is not identified by this design.

Plain-language reading. This is the most important negative finding in the paper, and its interpretation is under active investigation. “Pretraining prior” refers to the implicit population the model assumes when asked an English-language survey question with no country context—a population that looks American because the bulk of its training corpus is American English. On US items, adding synthetic US signal on top of that prior moves the policy away from the empirical US WVS distribution (TVD 0.396 → 0.526); the USA adapter also moves away from Mexico’s distribution (TVD 0.521 → 0.638), so the degradation is not specific to US items. The asymmetry with Mexico—where the matched adapter improves fit—is consistent with the prior-collision account (a weak base prior for Mexico versus a saturated one for the US), but competing explanations (adapter overfitting to the synthetic pair format, insufficient US-pair diversity) remain viable. The pre-registered persona-gradient protocol in the construct-validity layer is designed to adjudicate between these candidates; until it runs, the mechanism should be treated as unresolved. The practical takeaway that *structural preference estimation works where the base model does not already have an opinion* is a hypothesis, not an established conclusion.

Key Finding 3: Structural Under-Dispersion. Across all models, option log-likelihood scoring suffers from severe **under-dispersion** (dispersion bias ranging from -1.15 to -1.63). Softmax normalization concentrates probability mass on modal choices, underestimating real-world human population variance.

Plain-language reading. Imagine asking 1000 Mexicans a four-point trust question. Empirically, you might see 38% choose “trust a lot,” 42% “trust somewhat,” 15% “not very much,” and 5% “not at all.” The model, asked the same question, tends to give something like 68%, 27%, 4%, 1%. The modal answer is right; the tails are too thin. This is a known property of softmax scoring: because the model puts probability mass on the option it ranks highest and the temperature is fixed at 1, less-preferred options are exponentially down-weighted. Real populations are flatter—they contain contrarians, inattentive respondents, and translation ambiguity—and the LLM’s option scores do not capture that residual variance. The implication for empirical use is that any claim about *absolute shares* (e.g., “42% of Mexicans answer X”) is unsupported by the current pipeline; claims about *direction and rank* are.

5. Scope of Methodology & Reliable Applications

An honest reading of the methodology requires defining where the current pipeline is scientifically reliable versus where downstream deployment would be invalid.

5.1 Reliable Applications

1. **Implicit Preference Internalization for Non-WEIRD Populations.** *What this means:* the country-specific disposition shows up in the model’s choice probabilities without anyone naming the country in the prompt—the effect lives in the weights, not in the context window. *Why it matters:* persona prompting—the dominant approach in the prior literature—is brittle (it disappears when the chat ends) and cannot be aggregated across thousands of synthetic agents because each session is independent. A parameter-level encoding can be re-used, audited, and re-evaluated under controlled conditions.

2. **Relative / Ordinal Comparative Statics.** *What this means:* tracking *direction of change* in a preference dimension across populations, instruments, or experimental arms, not the absolute share. *Concrete use:* a researcher can compare how the *order* of trust, patience, or altruism shifts between two countries, or estimate whether a proposed intervention changes the implied reward margin on a given survey item, even when the exact percentage-share predictions are not credible.
3. **Pre-Fielding Instrument Stress-Testing.** *What this means:* before paying for expensive household survey fieldwork, a research team can load each country adapter and probe a draft questionnaire for items on which the model distributions diverge sharply from a target population benchmark. *Concrete use:* flagging survey items where the WVS reference distribution in the focal country disagrees with both the base and matched adapter is a cheap diagnostic for translation artifacts, response-scale mismatch, or genuine cultural divergence—useful whether or not the eventual deployer intends to use the model for inference.
4. **Zero-Shot Option-Probability Probing.** *What this means:* the log-probability scoring protocol is a clean diagnostic harness. It isolates the model’s *likelihood assignment* over structured choices from generation artifacts such as chat formatting, refusals, or surface fluency. *Concrete use:* when someone wants to ask “what would this model believe if forced to choose,” option-likelihood scoring answers that question without confounds.

5.2 Unreliable Applications & Boundaries

1. **Absolute Headcount Forecasting & Polling.** *What this means:* the pipeline does not support predictions of the form “X% of country *c*’s population will choose response *y*.” The softmax’s structural under-dispersion (Finding 3) compresses distribution tails and biases modal-option probability upward. *Why it matters:* an economist accustomed to using survey weights and stratified sampling to forecast election margins will not get the same kind of number from this pipeline—and that is by design, not by oversight.
2. **Fine-Tuning Over Already-Dominant Priors.** *What this means:* when the base model already encodes strong prior beliefs about a population (the WEIRD case), the DPO adapter *overshoots* the target distribution. *Why it matters:* the natural reaction would be to fine-tune more aggressively; the measured degradation suggests the right reaction may be to leave the base model alone or use a regularizer that anchors it—a hypothesis that Streams 1–2 are designed to test.
3. **Free-Form Conversational Persona Chat.** *What this means:* option-likelihood validation does *not* certify open-ended chat. A persona that gets the modal answer right on a structured questionnaire may still sound generically prosocial in free dialogue. *Why it matters:* downstream users wanting “an agent that speaks like someone from country *c*” in a chatbot context should not treat the option-likelihood validation as evidence about the chatbot experience.
4. **Granular Trait Elasticities on Thin Cells.** *What this means:* aggregate country-level metrics are credible; per-dimension per-question elasticities on small cells (e.g., Altruism with $n = 3$ items) carry wide confidence intervals and should not be quoted as calibrated structural parameters.

6. Proposed Next Steps: A Workplan

The Tier-1 and Tier-2 results jointly establish that the SCA 2.0 pipeline operates as a structural preference estimator for populations under-represented in pretraining, and that the nomological network it instantiates has at least one strongly attested link (convergent validity on Mexico) and one strongly informative negative case (the USA adapter’s degradation, whose mechanism remains unresolved). The agenda below is the *minimal credible next iteration* of the program—the steps we believe would either tighten the established result, repair the boundary failures, or move the methodology toward a paper-ready overidentification test.

We organize it into five streams, each with a primary deliverable, a rough effort estimate, and a falsification criterion (i.e., what result would tell us the stream is not worth continuing).

Stream 1: More countries, as a falsification test for prior-collision. The leading interpretation of the US failure is that fine-tuning over an already-dominant pretraining prior causes overshooting. The cleanest test of that interpretation is to add *another* country whose pretraining prior is weak but whose GPS profile differs from Mexico in a different direction. The natural candidate is Argentina: the GPS data place Argentina below the United States on patience, and Argentina is not a major pretraining-corpus country for an English-centric base model. If the MEX pattern replicates on ARG (matched > cross, base on ARG $\leq$ matched), we have evidence that the result is about *non-dominance* of the prior rather than about anything special about Mexico. If the ARG adapter overshoots despite a weak prior, the story is more complicated (perhaps contamination, perhaps language effects) and the pipeline needs work before scaling further. *Deliverable:* ARG adapter + Tier-2 evaluation on a small WVS-7 / AmericasBarometer trust core. *Effort:* roughly one week end-to-end (generation ~ 2 days, training ~ 1 day, evaluation ~ 1 day). *Falsification:* matched-vs-cross specificity on ARG items ≤ 0.5 .

Stream 2: A regularizer that prevents prior overshooting. If the prior-collision account is correct, the US overshooting has a structural cause that can be addressed without re-engineering the data pipeline. The DPO objective currently treats π_{ref} and π_{θ} symmetrically. A simple modification is to add a KL-anchoring term to the training loss that penalizes drift from π_{ref} *only in cells where the prior is already informative* (high-entropy reference scores on the target option set). The technical analogue in classical Bayesian updating would be a prior-strength-dependent weight on the likelihood. *Deliverable:* a modified DPOTrainer callback with a tunable anchor weight λ , evaluated on the US WVS items that currently fail. *Effort:* roughly three days (one engineer). *Falsification:* on US items, matched adapter $\text{TVD}_{\text{USA}} \geq \text{TVD}_{\text{base}}$ for all tested $\lambda > 0$. If even strong anchoring does not help, the implication is that *for populations with strong pretraining priors, the right object is the base model itself* and the program should narrow its scope to non-dominant priors.

Stream 3: Hyperparameter sensitivity for β and contamination. The current pipeline fixes $\beta = 0.1$ and applies no explicit control over contamination ratios (the share of the synthetic pair’s score gap attributable to non-target dimensions). Both are known drivers of DPO behavior: β controls how aggressively the policy moves away from the reference, and contamination is the gap between the synthetic training signal and the latent dimension we are claiming to estimate. A small grid over $\beta \in \{0.05, 0.1, 0.3\}$ and over training-set contamination thresholds $\in \{1.4, 2.0\}$ will tell us (i) whether the Mexico result is robust or whether it is a $\beta = 0.1$ artifact, and (ii) whether stricter contamination filtering improves out-of-sample WVS fit at the cost of training-set size. *Deliverable:* a 3×2 sensitivity table; an updated contamination threshold default in `synthetic_generation/config.py`. *Effort:* roughly one week (six full retraining runs plus evaluation). *Falsification:* Mexico matched-vs-cross specificity falls below 0.7 for any cell in the grid (then the result is fragile).

Stream 4: Overidentification test against an independent survey instrument. The strongest possible evidence that the adapter is learning the GPS dimension z rather than some unobserved correlate is to test the implied reward margins against an *independent* criterion that was not used in either training or the WVS evaluation. AmericasBarometer (AB), restricted to 2012–2019 trust-core items, is the natural choice: it is from a different survey house, with a different questionnaire tradition, and the AB items we have already merged into `data/merged/` are not present in the WVS-7 instrument. A simple overidentification test computes the implied marginal effect of a one-unit increase in the country’s patience score on the AB

trust-core TVD, with bootstrap standard errors, and asks whether the predicted direction matches what the GPS data alone would imply. *Deliverable*: a *J*-test-style diagnostic on AB trust core, reported in a new `eval_results_americasbarometer/` table that mirrors `eval_results_wvs_wave7/`. *Effort*: roughly one week (data prep ~ 1 day, scoring ~ 3 days, analysis ~ 1 day). *Falsification*: AB matched-vs-cross specificity ≤ 0.5 on items where WVS matched-vs-cross specificity was strong—this would suggest overfitting to the WVS questionnaire tradition rather than learning the latent z .

Stream 5: Calibrated absolute-share prediction. Finding 3 (structural under-dispersion) is currently the boundary that prevents any quantitative use of the pipeline. The simplest diagnostic fix is a per-question isotonic regression of model probabilities onto empirical probabilities, fit on a small calibration set held out from training. This is a standard post-hoc calibration step that does not require retraining. A more ambitious fix would be a temperature-controlled evaluation where $\tau > 1$ inflates tails and $\tau < 1$ sharpens them; reporting results across τ is a cheap and transparent way to bound the dispersion-bias problem. *Deliverable*: a calibrated option-likelihood protocol in `DPO_eval_WVS/` that supports both post-hoc calibration and a τ -sensitivity report. *Effort*: roughly three days. *Falsification*: after calibration, ECE on MEX WVS items does not drop below 0.06 (then calibration is a noise adjustment rather than a bias correction).

What we are explicitly *not* doing in the next iteration. We are not generating more synthetic scenarios for the same countries. The Tier-1 result on MEX is at the saturation point of the synthetic-pair signal (a doubling of the training set produced no measurable improvement in pilot runs, although we will report this systematically). We are not adding more GPS dimensions; the framework has a defensible theoretical prior on the six-dim Falk et al. set, and ad-hoc additions would compromise content validity. We are not switching base models, because the prior-collision result is a feature of the analysis (it tells us something about the base prior) rather than a bug to engineer around.

In summary, the highest-leverage next move is to expand the country panel to test the prior-collision hypothesis (Stream 1) and to add the AB overidentification test (Stream 4). Streams 2, 3, and 5 are engineering work that addresses known weaknesses; they should run in parallel because they touch disjoint parts of the codebase.

References

- [1] Bradley, R. A., and M. E. Terry (1952). “Rank Analysis of Incomplete Block Designs: I. The Method of Paired Comparisons.” *Biometrika* 39(3/4): 324–345.
- [2] Cao, Y., et al. (2023). “Assessing Cross-Cultural Alignment between ChatGPT and Human Societies.” *Proceedings of the First Workshop on Cross-Cultural Considerations in NLP*.
- [3] Cronbach, L. J., and P. E. Meehl (1955). “Construct Validity in Psychological Tests.” *Psychological Bulletin* 52(4): 281–302.
- [4] Capra, C. M., A. Gonzalez-Bonorino, and M. Pantoja (2025). “LLMs Can Model Non-WEIRD Populations: Experiments with Synthetic Cultural Agents.” Working paper; forthcoming, *Review of Experimental Economics*.
- [5] Aher, G., R. Arriaga, and A. Kalai (2023). “Using Large Language Models to Simulate Multiple Humans and Replicate Human Subject Studies.” *ICML*.
- [6] Durmus, E., et al. (2024). “Towards Measuring the Representation of Subjective Global Opinions in Language Models.” *Conference on Language Modeling (COLM)*.

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- [7] Falk, A., A. Becker, T. Dohmen, B. Enke, D. Huffman, and U. Sunde (2018). “Global Evidence on Economic Preferences.” *Quarterly Journal of Economics* 133(4): 1645–1692.
 - [8] Henrich, J., S. J. Heine, and A. Norenzayan (2010). “The Weirdest People in the World?” *Behavioral and Brain Sciences* 33(2–3): 61–83.
 - [9] Horton, J. J. (2023). “Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus?” NBER Working Paper 31122.
 - [10] McFadden, D. (1974). “Conditional Logit Analysis of Qualitative Choice Behavior.” In *Frontiers in Econometrics*, ed. P. Zarembka. Academic Press.
 - [11] Rafailov, R., A. Sharma, E. Mitchell, S. Ermon, C. D. Manning, and C. Finn (2023). “Direct Preference Optimization: Your Language Model is Secretly a Reward Model.” *NeurIPS*.